PRIME DIVISORS AND THE COMPONENTS OF AN EFFECTIVE CARTIER DIVISOR IN TERMS OF THE RELATIVE DIMENSION FUNCTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The definition of a prime divisor, of a dimension function, and the local shape of an effective Cartier divisor. Divisors, labels divisors-definition-Weil-divisor (proof line 2508), divisors-lemma-effective-Cartier-in-points (proof line 2515); Topology, label topology-definition-dimension-function (proof line 2510); Algebra, label algebra-lemma-one-equation (proof line 2518); Properties, label properties-lemma-codimension-local-ring (proof line 2519). The definitions give part (1); for part (2) the local equation of the divisor and the one-equation lemma give that the local ring at the generic point has dimension one, whence the dimension function drops by one.

Source reference. Original lemma 00KW: sources/stacks-chow/tags.txt, source lines 766–766. Navigation pointer for \ref{algebra-lemma-one-equation} used inside the counted span (line 2518); the argument that consumes it is retained in the span.

Source reference. Original definition 0BE2: sources/stacks-chow/tags.txt, source lines 14335–14335. Navigation pointer for \ref{divisors-definition-Weil-divisor} used inside the counted span (line 2508); the argument that consumes it is retained in the span.

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Source reference. Original lemma 0AG8: sources/stacks-chow/tags.txt, source lines 13186–13186. Navigation pointer for \ref{divisors-lemma-effective-Cartier-in-points} used inside the counted span (line 2516); the argument that consumes it is retained in the span.

Source reference. Original lemma 02IZ: sources/stacks-chow/tags.txt, source lines 3318–3318. Navigation pointer for \ref{properties-lemma-codimension-local-ring} used inside the counted span (line 2519); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2490); the argument that consumes it is retained in the span.

Source reference. Original definition 02I9: sources/stacks-chow/tags.txt, source lines 3292–3292. Navigation pointer for \ref{topology-definition-dimension-function} used inside the counted span (line 2510); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Assume X is integral.*

(1) *If $Z \subset X$ is an integral closed subscheme, then the following are equivalent:*
   (a) *Z is a prime divisor,*
   (b) *Z has codimension 1 in X, and*
   (c) $\dim_\delta(Z) = \dim_\delta(X) - 1$.

(2) *If Z is an irreducible component of an effective Cartier divisor on X, then* $\dim_\delta(Z) = \dim_\delta(X) - 1$.

Proof. Part (1) follows from the definition of a prime divisor (Divisors, Definition 0BE2) and the definition of a dimension function (Topology, Definition 02I9). Let $\xi \in Z$ be the generic point of an irreducible component Z of an effective Cartier divisor $D \subset X$. Then $\dim(\mathcal{O}_{D,\xi}) = 0$ and $\mathcal{O}_{D,\xi} = \mathcal{O}_{X,\xi}/(f)$ for some nonzerodivisor $f \in \mathcal{O}_{X,\xi}$ (Divisors, Lemma 0AG8). Then $\dim(\mathcal{O}_{X,\xi}) = 1$ by Algebra, Lemma 00KW. Hence Z is as in (1) by Properties, Lemma 02IZ and the proof is complete. □